

CHE-110
unit-1
(3 lectures)

Introduction and sustainable development :
introduction to environment, components of environment,
spheres of earth, multidisciplinary nature, scope and
importance of environmental studies, concept of
sustainability, sustainable development, carrying capacity,
pillars of sustainability, sustainable development goals

- An Environment is everything that is around us,
- which includes both living and nonliving things such as soil, water, animals and plants, which adapt themselves to their surroundings.
- It is nature's gift that helps in nourishing life on Earth.
- The environment plays an important role in the existence of life on the planet earth.
- The word Environment is derived from the French word “Environ” which means “surrounding.”

Environment is thus defined as “the sum total of water, air and land and the inter relationships that exist among them and with the human beings, other living organisms and materials.”

The scope and importance of environmental studies encompass several key aspects:

Multidisciplinary Approach: Environmental studies integrate principles from physical sciences, social sciences, and economics to address complex environmental issues.

Awareness and Conservation: They raise awareness about the critical need to safeguard and conserve our planet, highlighting the detrimental effects of human activities.

Interrelationships: The field examines the interactions between organisms and their environment, which is essential for understanding ecological balance and sustainability.

Practical Applications: Environmental studies bridge theoretical knowledge with practical applications, demonstrating the benefits of a well-managed environment.

These elements underscore the significance of environmental studies in fostering a sustainable future.



The Heart and Soul of the **Narmada Bachao Andolan**

Medha Patkar



AJENDRA SINGH

—
INDIA'S WATERMAN





1. Jadav Payeng Also known as the ‘forest man of India,’ Jadav Payeng comes from the state of Assam.

2. Sumaira Abdulali

3. Salim Ali
Also known as the ‘Birdman of India,’

4. M. S. Swaminathan
Known as the ‘Father of India’s Green Revolution,’

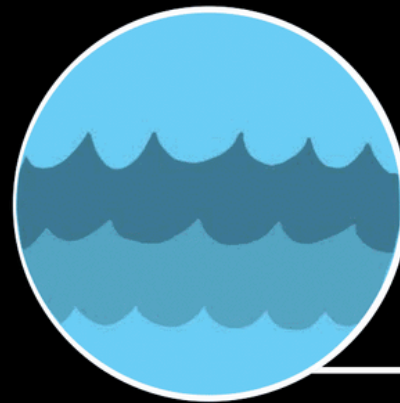
EARTH LAYERS AND SPHERES

BIOSPHERE



The biosphere is the zone encircling the Earth where life exists and thrives. Scientists have found things living deep within the Earth's crust, and high up in the atmosphere. Thus, the biosphere overlaps all three other spheres.

HYDROSPHERE



2/3rds of the Earth's surface is covered with water. This water makes up the hydrosphere. The word 'hydro' means water.. The hydrosphere includes oceans and seas, as well as lakes, ponds, rivers, streams, and swamps across the Earth.

LITHOSPHERE

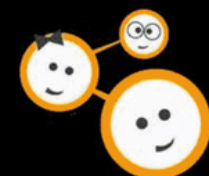
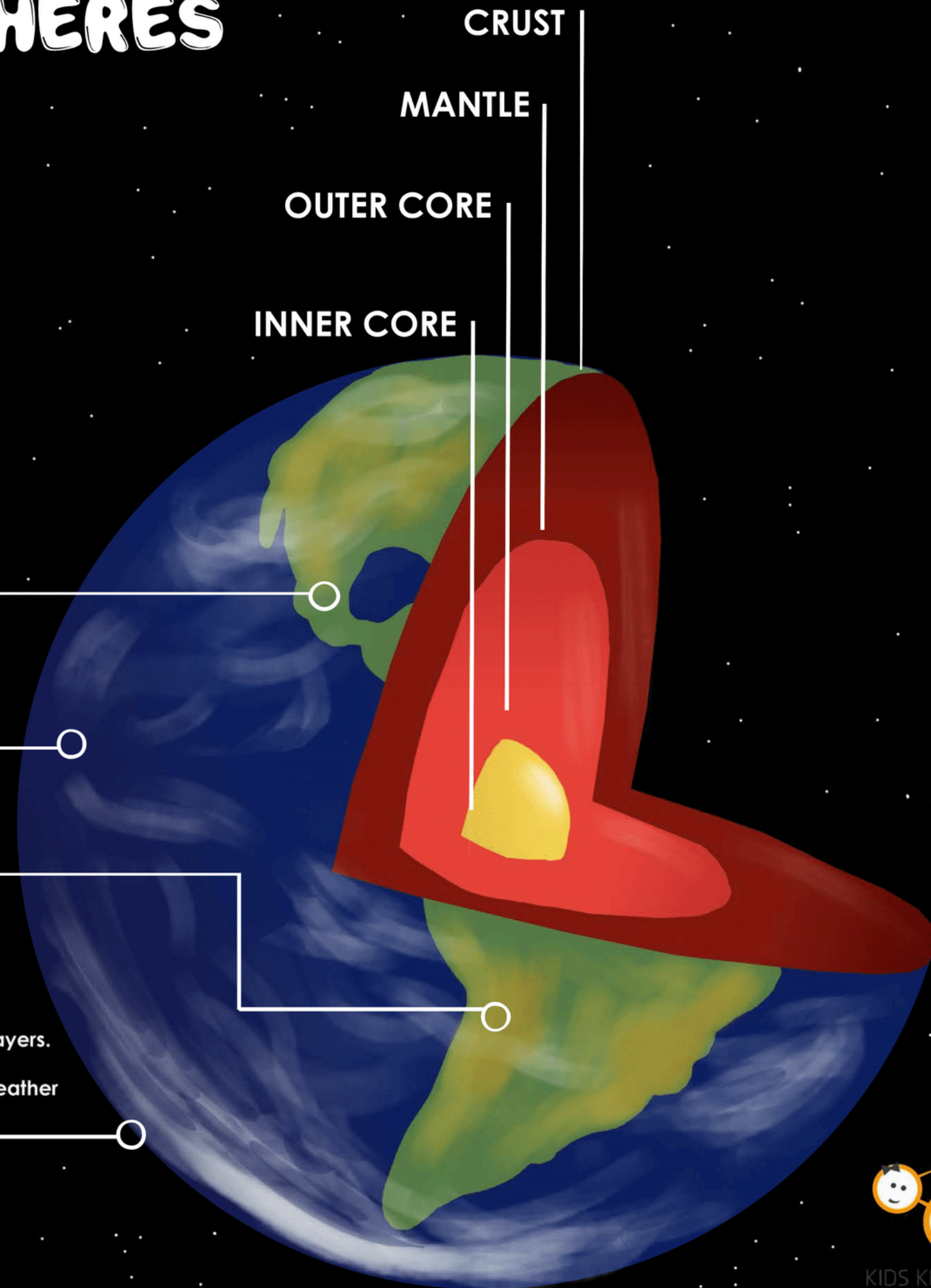


The lithosphere is made up of all the rocky materials on Earth. This includes the land, dirt, mountain, and rocks found at the surface as well as the materials found deep within its interior.

ATMOSPHERE



Sphere of gas the encircles the Earth. The atmosphere itself is made up of many layers. These layers are divided up by the types of gases, temperatures, and weather that takes place in each.



Components of Environment

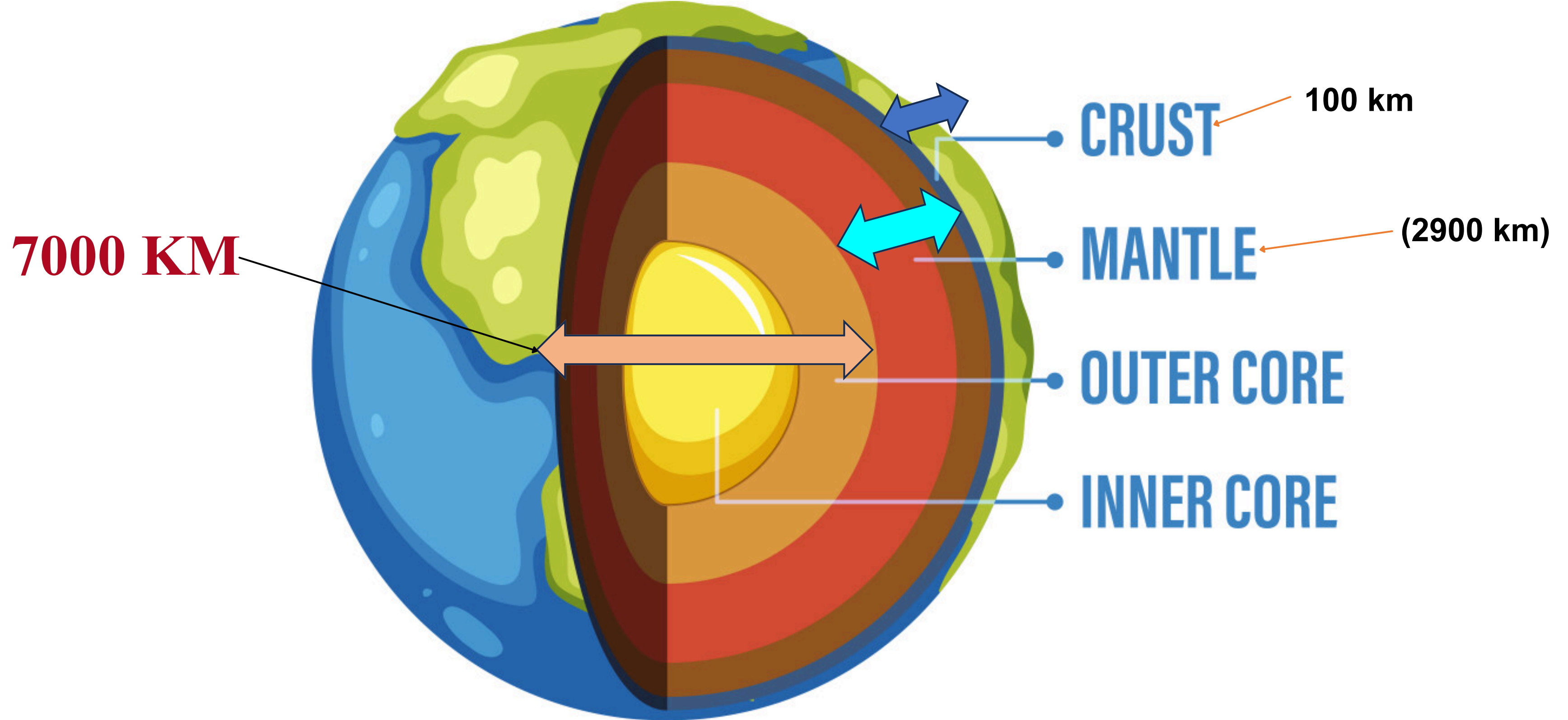
The four major components of the environment are

- lithosphere (rocks),
- hydrosphere (water),
- atmosphere (air), and
- biosphere (life):



Lithosphere

- The term 'Lithosphere' is derived by combining the Greek word "lithos," which signifies rock, with the word "sphere."
- The lithosphere is the solid outer portion of Earth which includes Earth's crust as well as the underlying cool, dense, rigid uppermost portion of the upper mantle.
- At the center of the Earth is a structure called the Core, which has a diameter of around 7000 km.
- Surrounding the core is the 2900 km thick mantle. The lithosphere extends from the surface of Earth to a depth of approximately 100 km. The tectonic plates of Earth make up the majority of the lithosphere.

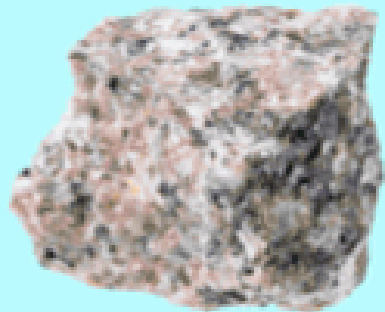


LAYERS OF THE EARTH

Types of Rocks

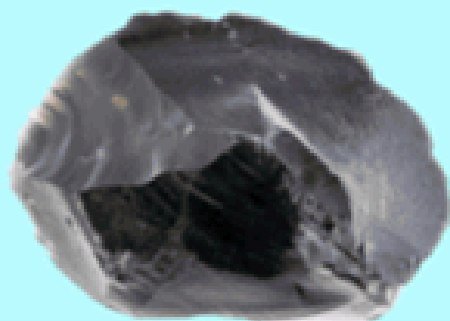
Igneous

- Forms from magma or lava solidification
- Hard, no layers



Granite

Intrusive
slow magma cooling



Obsidian

Extrusive
rapid lava cooling

Sedimentary

- Forms from sediment compaction
- Crumbly, layered



Sandstone

Clastic
compacted broken rocks



Limestone

Chemical
compacted dissolved minerals



Coal

Organic
compacted biogenic matter

Metamorphic

- Forms by transformation of other rocks
- Relatively hard, may or may not have layers



Slate

Foliated
has layers



Marble

Non-Foliated
no layers

Hydrosphere

- It includes all types of water bodies on Earth, such as freshwater (rivers, lakes, ponds, streams), marine (oceans, seas), and groundwater resources, among others.
- There are three states of water: gas (water vapour), liquid (water), and solid (ice).
- It occupies 71% of the planet's surface.
- The oceans contain 97.5% of all the water on Earth, which is salt water. Freshwater makes up only 2.5% of all water on Earth.
- Of total, 68.9% is in frozen forms, such as glaciers, while 30.8% is accessible as groundwater.
- A readily available quantity of 0.3% can be found in lakes, reservoirs, and rivers.

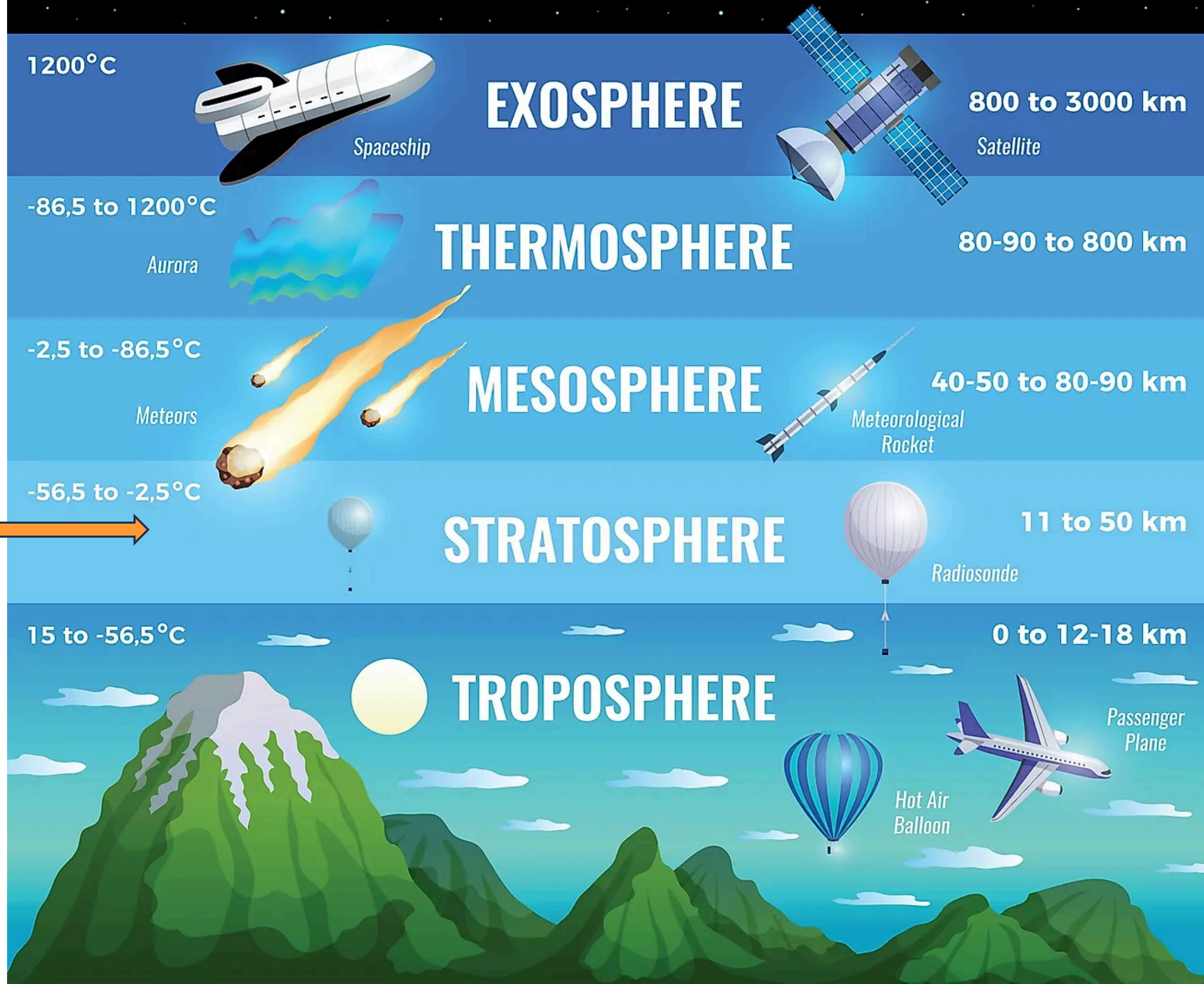
Atmosphere

- The earth is enveloped by thick gaseous layers that make up the atmosphere. It stretches up to 300 km.
- Life is supported by the atmosphere which contains a lot of oxygen. It is composed of gases such as nitrogen (78.08%), oxygen (20.95%), argon (0.93%), carbon dioxide (0.039%), and trace amounts of hydrogen, helium, and noble gases.
- The troposphere, stratosphere, mesosphere, thermosphere, and exosphere are the layers that rise above the surface of the earth.

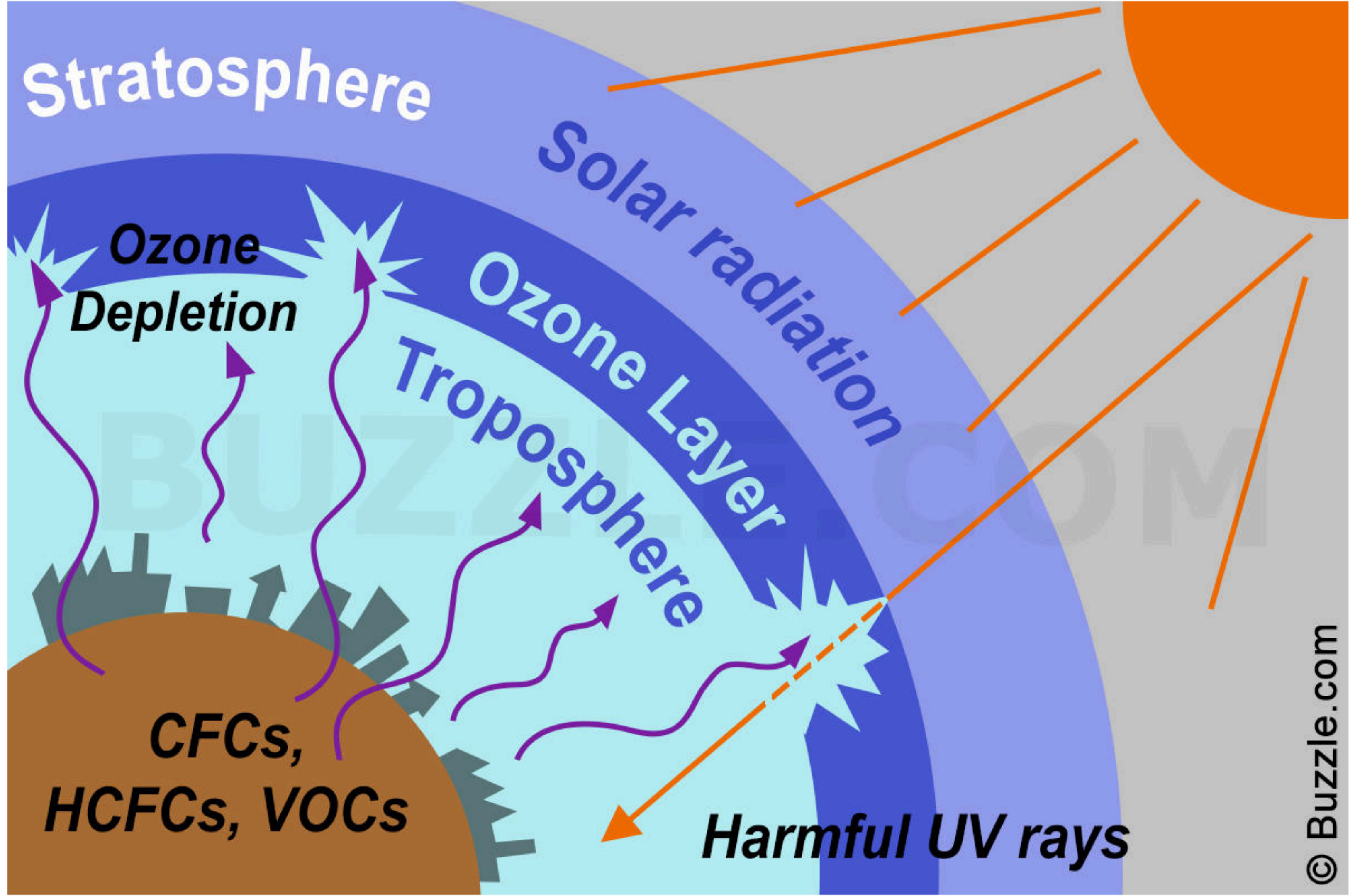
- **The lowest layer is called the troposphere.**
- **It rises from ground level to a height of roughly 11 kms above sea level.**
- **The stratosphere is the next layer above.**
- **It stretches from the top of the troposphere to about 52 km above the ground.**
- **The ozone layer is present in stratosphere.**
- **Ozone molecules in this layer absorb ultraviolet light from the sun and convert the UV energy into heat.**

- **The mesosphere is located above the stratosphere.**
- **It rises to a height of approximately 85 kms above Earth.**
- **The mesosphere is where most meteors burn up.**
- **The thermosphere is the layer of extremely rare air above the mesosphere.**
- **The thermosphere absorbs UV and high energy X-rays from the Sun, which can cause it to heat up to thousands or even hundreds of degrees. The top thermosphere can have temperatures as high as 2,000°C.**

Layers of Earth's Atmosphere



Ozone layer →



concept of sustainability,
sustainable development

**“meeting the needs of the
present without
compromising the ability
of future generations to
meet their own needs”**

Examples of Environmental Sustainability

1. Using renewable
energy

2. Limiting waste
and pollution

3. Conserving water

4. Using energy-efficient
office equipment

5. Reducing travel and
travel emissions

6. Creating products that
are easy to recycle

7. Constructing
and maintaining
sustainable buildings



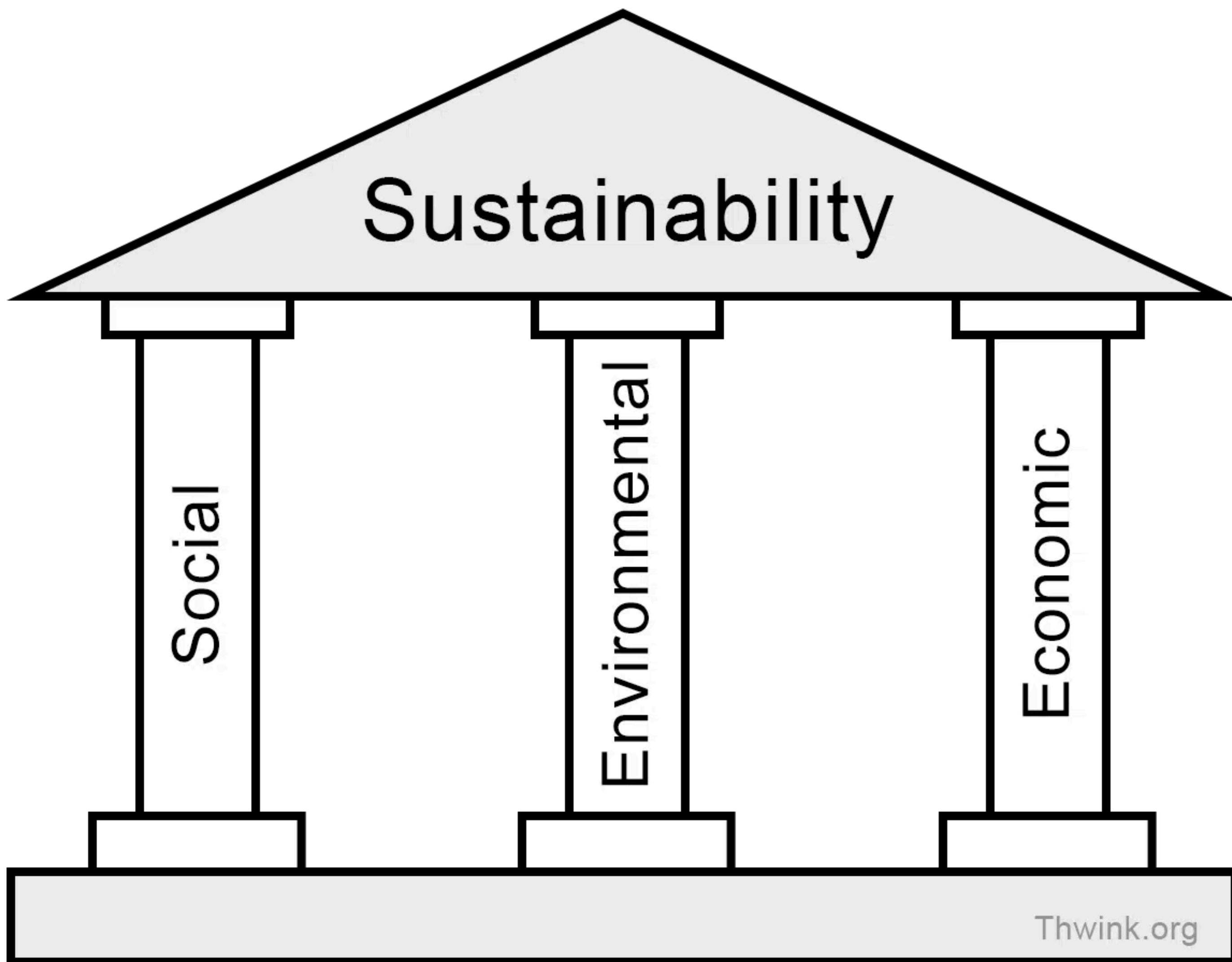
The ‘Three Pillars of Sustainability’ is another popular framework used to describe what sustainable development is.

This tool conveys that sustainability consists of three factors

1. environmental,

2. social, and

3. economic

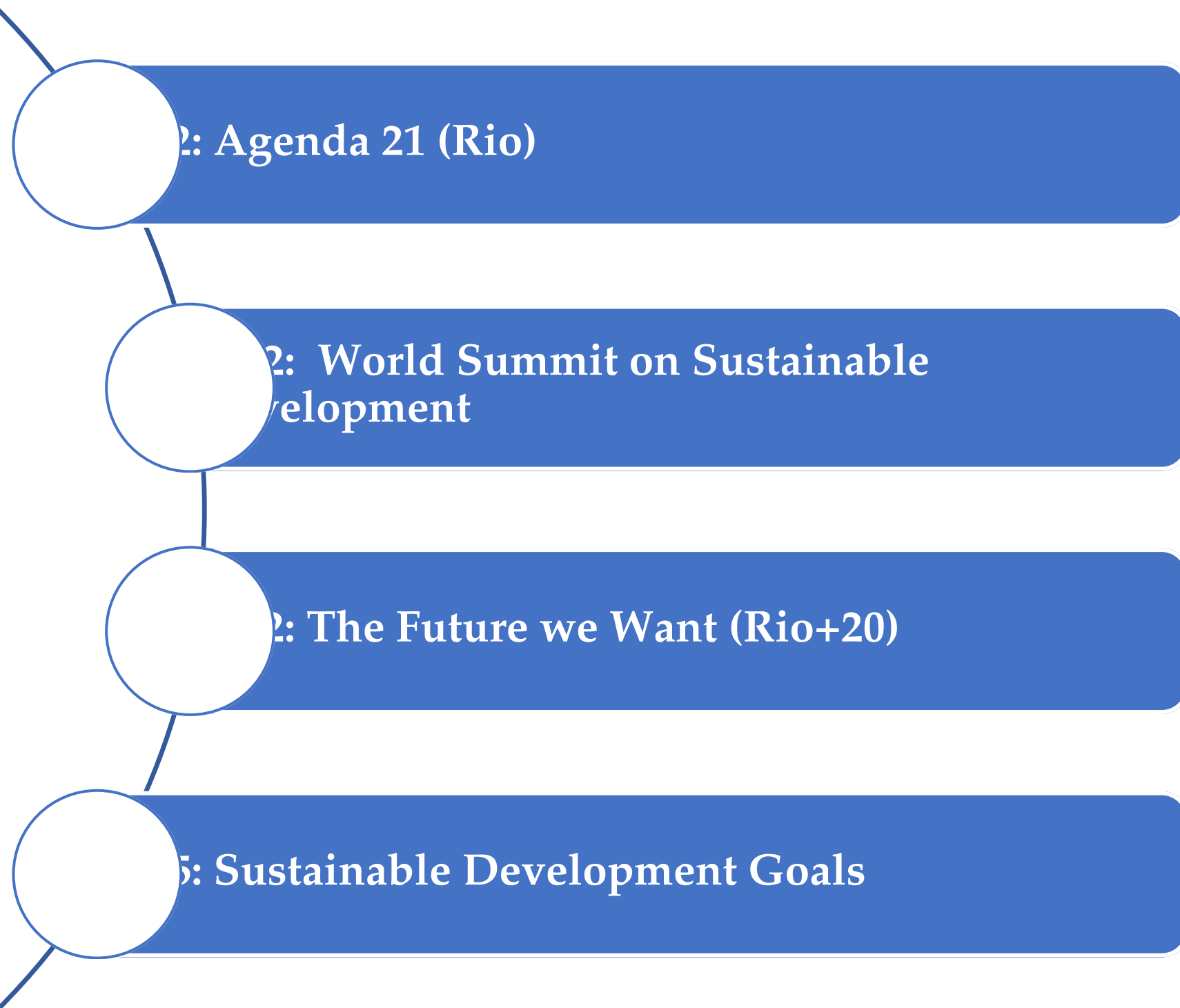


Environmental sustainability is perhaps the most obvious of the three pillars, as it symbolizes the importance of things like natural resources and biodiversity to support life on Earth.

Social sustainability places importance on social structures, well-being, and harmony; all factors that poverty, wars, and injustices can affect.

Economic sustainability describes the ability of an economy to grow. This is especially important in today's societies, at a time when many sustainable initiatives require financing and a strong economic rationale.

The political process





From unsustainable to sustainable development

- Does it protect our biodiversity?
- Does it prevent soil erosion?
- Does it slow down population growth?
- Does it increase forest cover?
- Does it cut off the emissions of CFC, SO_x, NO_x and CO₂ ?
- Does it reduce waste generation and does it bring benefits to all?

Measures for Sustainable Development

- Using appropriate technology
- Reduce, Reuse, Recycle approach
- Prompting environmental education and awareness
- Resource utilization as per carrying capacity
- Implementing effective planning for Population Control
- Less dependence on non-renewable natural resources

Sustainable Environmental Technology



Solutions for
natural resource
depletion

environmental
monitoring and
resource
management

Environmental
policies and
framework

Solid waste and
e-waste
management

Recovery of
metals and other
beneficial

Biofertilizer and

Sustainable
materials for the
cleanup of toxic

Waste-to-energy
conversion (bio-
CNG, bioethanol
biofuel, etc.)

Minimization of
greenhouse
emissions and
other effects of
climate change

The key aspects for sustainable development

- Inter-generational equity
 - minimize any adverse impacts on resources and environment
 - stop over-exploitation of resources, reduce waste discharge and emissions and maintain ecological balance
- Intra-generational equity
 - minimize the wealth gaps within and between nations
 - The technology should address to the problems of the developing countries
 - producing drought tolerant varieties for uncertain climates
 - vaccines for infectious diseases
 - clean fuels for domestic and industrial use

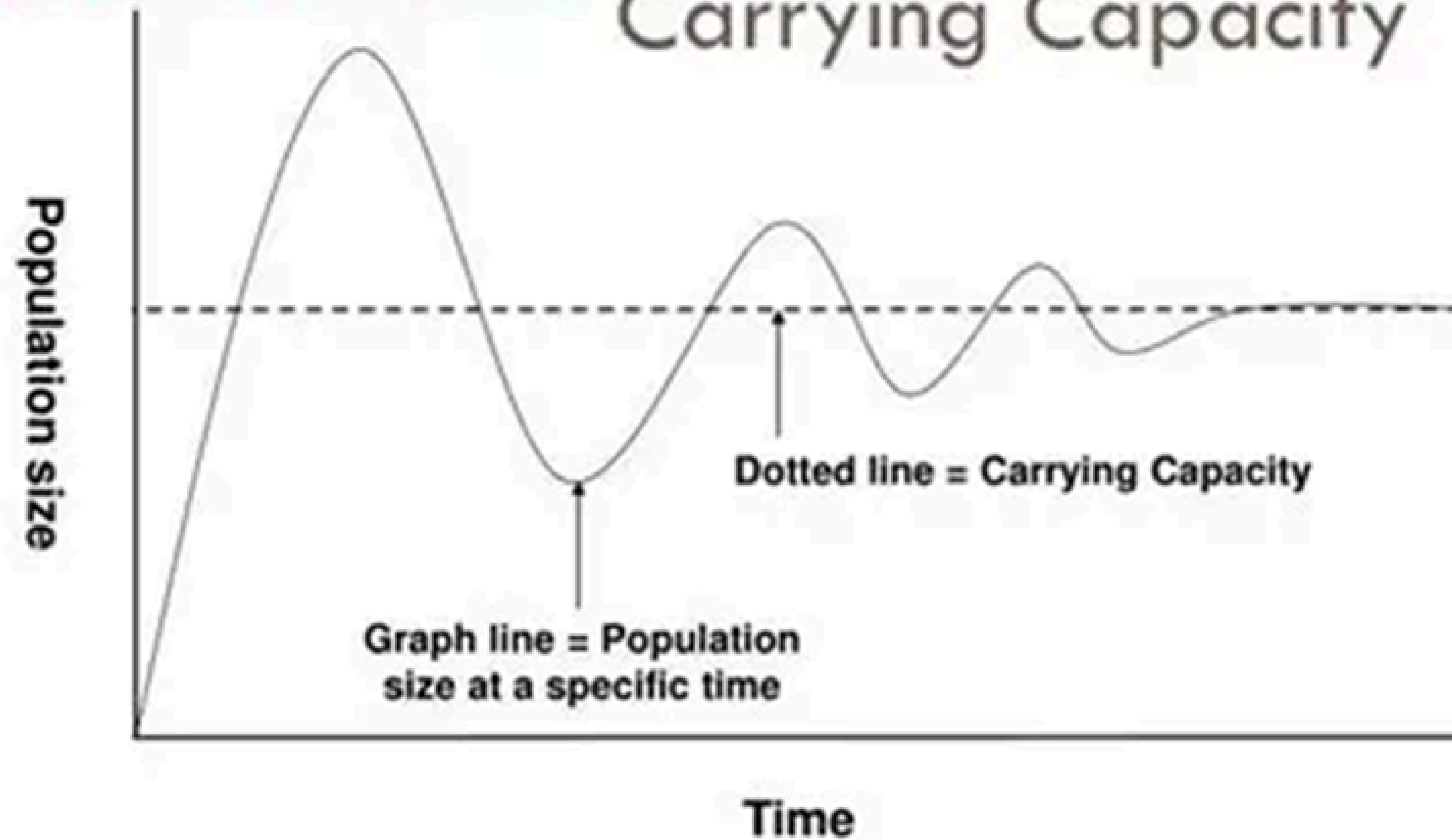
- **Carrying Capacity (k):**

- The maximum population size that can be supported by the available resources
- There can only be as many organisms as the environmental resources can support

- Until eventually, the population size **BECOMES STABLE AT THE CARRYING CAPACITY**
- Birth rate = death rate
- When a population is **BELOW** its carrying capacity, it will **INCREASE** in size
- Birth rate exceeds death rates



Carrying Capacity



- **The carrying capacity of an ecosystem is the maximum population size of a biological species that can be sustained by that specific environment, given the food, habitat, water, and other resources available.**
- **Carrying capacity of the environment implies that the resources extraction is not above the rate of regeneration of the resources and the wastes generated are within the assimilating capacity of the environment.**

- **Carrying capacity is applied to the maximum population an environment can support in ecology, agriculture and fisheries.**
- **The term carrying capacity had been applied to a few different processes in the past before finally being applied to human population limits in the 1950s.**
- **The notion of carrying capacity for humans is covered by the notion of sustainable population.**

17 goals under the Sustainable Development Goals

- 1. End poverty in all its forms everywhere**
- 2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture**
- 3. Ensure healthy lives and promote well being for all at all stages**
- 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all**
- 5. Achieve gender equality and empower all women and girls**
- 6. Ensure availability and sustainable management of water and sanitation for all**

17 goals under the Sustainable Development Goals

8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

9. Built resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

10. Reduce inequalities within and among countries

11. Make cities and human settlements inclusive, safe, resilient and sustainable

12. Ensure sustainable consumption and production pattern

13. Take urgent actions to combat climate change and its impact

17 goals under the Sustainable Development Goals

14. Conserve and sustainably use the oceans, seas and marine resources

15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably managed forests, combat desertification and halt and reverse land degradation and halt biodiversity loss

16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels

17. Strengthen the means of implementation and revitalize the global partnership for sustainable development

How are countries performing?



there is a dashboard for that.....

<https://dashboards.sdgindex.org/rankings>

SDG Indicators



Rankings

The overall performance of all 193 UN Member States



dashboards.sdindex.org/rankings



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Rank	Country	Score	Performance by SDG
1	 Finland	87.02	
2	 Sweden	85.74	
3	 Denmark	85.26	
4	 Germany	83.67	
5	 France	83.14	
6	 Austria	83.01	
7	 Norway	82.72	

The overall performance of all 193 UN Member States

[OVERALL](#)
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Countries are ranked by their overall score. The overall score measures the total progress towards achieving all 17 SDGs. The score can be interpreted as a percentage of SDG achievement. A score of 100 indicates that all SDGs have been achieved.

Click on a country for details.

Rank

Country

Score

Performance by SDG

99

 India

66.95



